



RetailMind

Semantic Product Knowledge
AI
Engine

B.Tech Project
Computer Science & Engineering

The Search Gap



Keyword Failure

Traditional systems only match exact words, missing the actual intent behind user searches.



NL Query Need

Modern users expect to ask questions like "Something warm for winter hiking" rather than typing keywords.



Information Gap

Difficult to retrieve deeply relevant product specs hidden within unstructured product descriptions.

RetailMind AI: The Solution

Beyond Keywords

- ✓ **Semantic Search:** Understands context and word meanings through embeddings.
- ✓ **RAG Integration:** Retrieves specific catalogue data to augment LLM answers.
- ✓ **Precision:** Delivers meaningful product suggestions with high relevance scores.



Project Vision



Intelligence

Build a system that thinks like a retail assistant, not a database filter.



Real Dataset

Engineered to work on diverse Shopify product catalogues with varied metadata.



Scalable

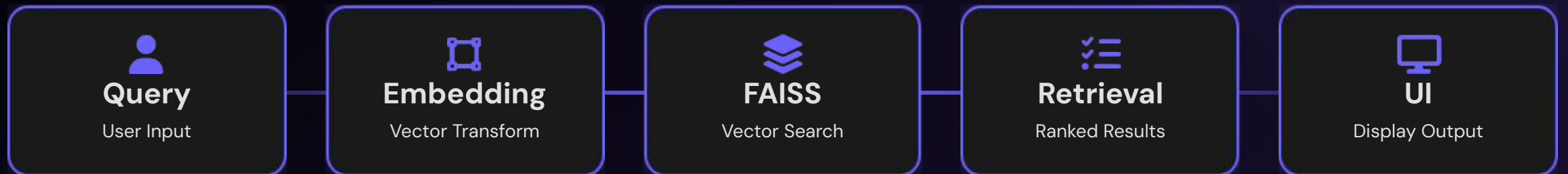
Utilizes vector indexing to handle thousands of products with millisecond latency.



User Centric

Simplifies the shopping journey from query to checkout via intuitive UI.

Architecture Flow



Technology Stack



Backend

Flask API for lightweight request handling and model serving.



Vector DB

FAISS (Facebook AI Similarity Search) for optimized CPU/GPU indexing.



Embeddings

Sentence Transformers (SBERT) for high-dimensional semantic mapping.



Frontend

Modern HTML5, CSS3, and JS for an interactive chat-first UI.

rjac/e-commerce -customer-support-qa

48k+
Total FAQ'S

5000
Processed Sub-set

Feature	Description	Type
Customer Query	User's question describing the issue or problem	Textual
Support Answer	Official response used to resolve the query	Textual
Issue Category	High-level classification of the problem (e.g., Returns, Shipping)	Categorical

FAISS Vector Indexing

100x
Search Acceleration

Traditional search compares query to every item (Linear $O(N)$). FAISS uses space partitioning for sub-linear search speeds.

Query: "Red shoes"
Vector: [0.12, -0.45, ..., 0.89]
FAISS → Match ID: 4029



Why Index?

- Avoid slow brute-force scans.
- Maintain constant response time as data grows.
- Optimized memory management.

HNSW Graph Index

Hierarchical Navigable Small World

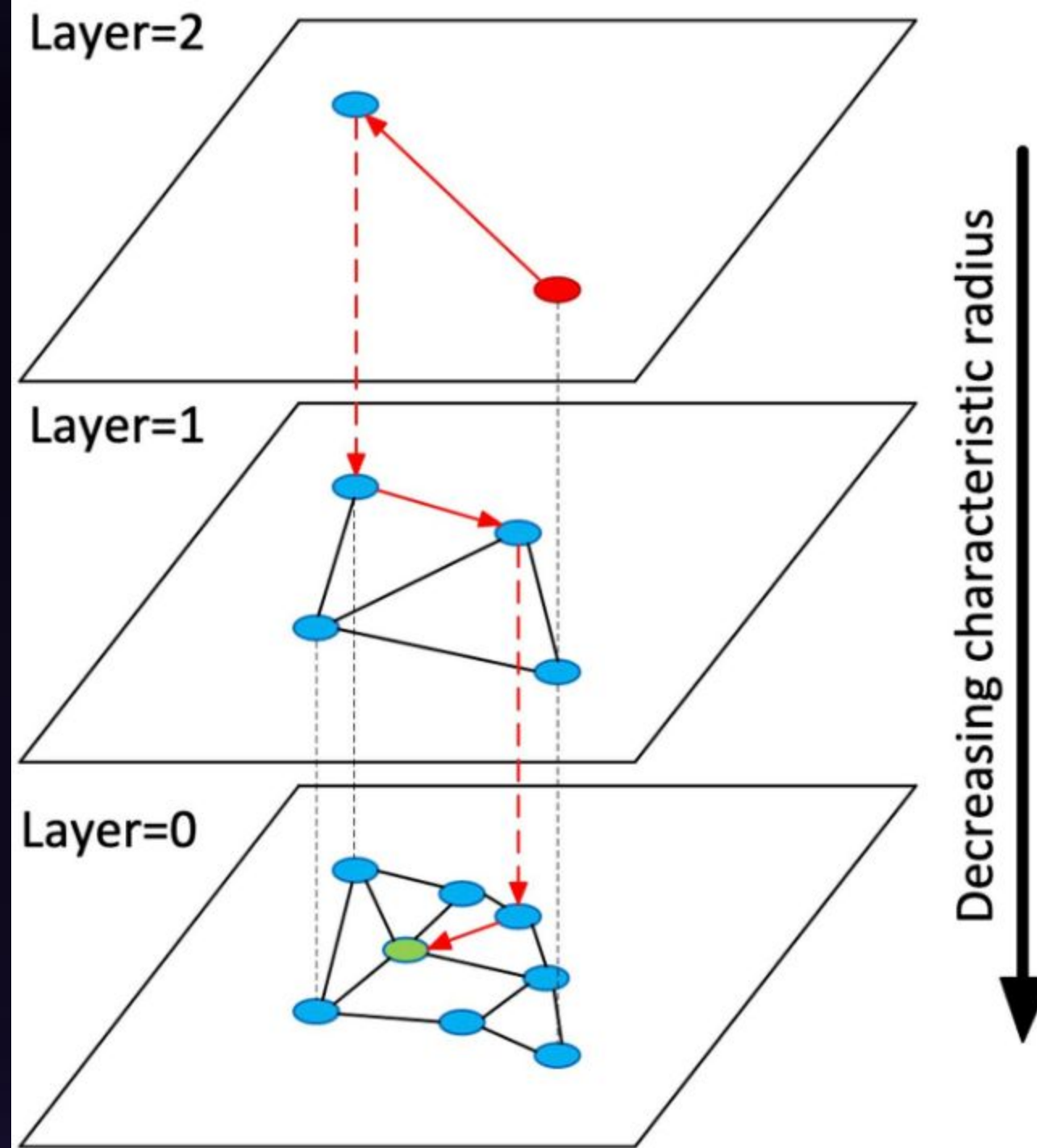
Multi-layer structure for fast routing.

Graph-based "proximity" search.

Hyperparameters:

ef_{Search} , $ef_{Construction}$

Balancing search speed vs. accuracy/recall.



RAG Workflow



1. Retrieve

FAISS finds the top-K relevant products based on query embeddings.



2. Augment

Combine product info (price, features) with the original query.



3. Generate

Response engine produces a natural, conversational product summary.

Ensures the AI never "hallucinates" non-existent products.

Output:



RetailMind AI

CUSTOMER SUPPORT KNOWLEDGE ENGINE

⊕ HNSW Index

● Ready · 0 docs

SAMPLE QUERIES

What is the return policy for electronics?

How do I claim warranty for a defective product?

Can I cancel my order after it has shipped?

My product is not compatible with my device

How long does a refund take to process?

Product damaged during delivery, what to do?

How to change my delivery address after ordering?

Payment failed but amount was deducted

TOPICS COVERED

Returns & Refunds

Warranty Claims

Order Cancellation

R

Hello! I'm **RetailMind AI** — your intelligent customer support knowledge engine.

I can answer questions about **warranty coverage**, **return conditions**, **compatibility details**, **order cancellations**, **refund timelines**, and more — all retrieved semantically from a pre-indexed support knowledge base.

Ask about warranty, returns, compatibility, shipping, or any product support query...



Ctrl+Enter to send ·

Output:



RetailMind AI

CUSTOMER SUPPORT KNOWLEDGE ENGINE

⊕ HNSW Index

● Ready · 0 docs

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Returns & Refunds

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Order Cancellation

Shipping & Delivery

Compatibility

Payment Issues

Account Access

Product Specs

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U

What is the return policy for electronics?

R

Found 5 relevant support documents for: "What is the return policy for electronics?"

1. Products not eligible for returns

Area: Cancellations and returns · Category: Replacement and Return Process · Product: Laptop

Complexity: less · Relevance: Excellent match (0.83)

Answer: Although returns for opened laptops are not allowed, the agent offered troubleshooting and provided manufacturer's warranty assistance, which led to the customer getting the laptop repaired or replaced by the manufacturer.

2. Returns after the specified time period in the seller's Returns Policy

Area: Cancellations and returns · Category: Replacement and Return Process · Product: Speaker

Complexity: medium · Relevance: Excellent match (0.82)

Ask about warranty, returns, compatibility, shipping, or any product support query...

Ctrl+Enter to send



Output:

- How do I claim warranty for a defective product?
- Can I cancel my order after it has shipped?
- My product is not compatible with my device
- How long does a refund take to process?
- Product damaged during delivery, what to do?
- How to change my delivery address after ordering?
- Payment failed but amount was deducted

TOPICS COVERED

- Returns & Refunds
- Warranty Claims
- Order Cancellation
- Shipping & Delivery
- Compatibility
- Payment Issues
- Account Access
- Product Specs

R

Found 5 relevant support documents for: "Payment failed but amount was deducted"

1. Billing Discrepancies/Overcharging

Area: Order · Category: Invoice and Payment · Product: Food Processor

Complexity: medium · Relevance: Strong match (0.74)

Answer: Apologized for the billing error, confirmed the overcharge, and assured a refund within 3-5 business days to the original payment method. There were no further questions or concerns.

2. Checks performed for returned items

Area: Cancellations and returns · Category: Return Checks and Fees · Product: Tablet

Complexity: less · Relevance: Strong match (0.74)

Answer: I had to deduct a restocking fee of \$50 from your refund because the returned tablet was damaged after inspection.

U

Payment failed but amount was deducted

Ask about warranty, returns, compatibility, shipping, or any product support query...

Ctrl+Enter to send

Project Summary

- ✓ Successfully implemented Semantic Search with FAISS & SBERT.
- ✓ Reduced search friction using HNSW graph indexing for sub-second recall.
- ✓ **Future Scope:** Integration with Voice Search and Multi-lingual support.

Thank You!

Any Questions?